

Draft - expert review pending

This unit is complete and passes the platform's automated checks, but no reviewer has read it yet. Treat its sources and claims with care.

Unit 3 teacher notes

Sequencing note

Unit 3 runs outside-in. Topic 3.1 gives the OS's job description and types; 3.2 opens its two everyday offices (files and processes); 3.3 follows its traffic management (input and output); 3.4 descends to the kernel that owns all three. Teach in order, and keep returning to the same two classroom scenes: the clerk's lost letter (files) and the afternoon slowdown (processes). They carry the unit.

Common misconceptions to probe

1. **"The operating system is just the desktop picture and the taskbar."** Probe before revealing Topic 3.1: ask who is in charge before any app opens. Follow the boot sequence from Unit 2 and name the OS as the always-present layer.
2. **"Closing a window stops the program."** The most persistent one. Probe with the process list on a real machine: close a window, watch the process survive, then end the process and watch what vanishes.
3. **"A slow computer is an old computer."** Probe with the time-sharing idea: what happens to each program's slice when more programs demand turns? Age is one cause; crowding is the commoner one.
4. **"Deleted means erased."** Probe after Topic 3.2: what does the OS actually remove when you delete a file? (Its index record; the space is marked reusable.)
5. **"The internet enters the computer by magic."** Carried over from Unit 2; Topic 3.3's chain gives it names again - interface, driver, OS.

Managing the activities

Topic 3.1 - Function spotting (20 minutes, pairs). The trap is naming the visible action (printing) instead of the OS function underneath (input/output management). Require the sentence "the OS did X" for every item.

Topic 3.2 - The naming rescue and the process list (20 minutes, pairs). The naming half is deliberately concrete; the process-list half is the eye-opener. If machines are available, have pairs find one background process and one visible application, and close nothing without permission. The swap-and-find test at the end is the assessment.

Topic 3.3 - The diagnostic ladder (20 minutes, pairs). The trap is jumping to the device ("the printer is broken"). Insist every diagnosis names the link and the evidence. The five faults are ordered roughly from easy to arguable; let the arguments happen.

Topic 3.4 - Layer sort (15 minutes, pairs or threes). The driver card is the intended argument: it is software, but it serves the kernel's device control. Do not resolve it early; the reasoning is the point.

Working with the guide's strategies

Lectures suit the two structural pictures: the OS-types table (3.1) and the four layers (3.4). Keep each to the figure and one example. **Interactive discussion** and **question-answer sessions** fit the misconception probes; run them before each correction. **Demonstrations** are the unit's heart: open the process list live, delete a file and discuss what happened, plug in a USB drive and narrate the chain, print two documents and watch the queue. **Case study analysis** is every diagnostic scenario in 3.3 and the two ERQ fault cases; treat the afternoon slowdown as a standing case the class returns to.

Practical work and practicum links

- **Hands-on computer exercises:** the process list, the file-scheme design against the school's real document store, and the diagnostic walk-through of one real fault.
- **Individual assignments:** the office rescue plan (3.2) and the print-fault analysis (3.3).
- **Group assignments:** the naming rescue, the diagnostic ladder and the layer sort.
- **Internet research tasks:** optional - have trainees find the name of the OS their own phone runs and one fact about its type; require a source.
- **Presentations:** the staff-meeting explainer (3.4) presents well; the layer sort can end with each group defending one placement.

The practicum tasks (watching the office computer, the document store sketch, the fault story) extend the school ICT profile begun in Unit 2 and feed Unit 4's security audit.

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